QUIZ 1 NETWORKING

The correct option is:

MANs are typically operated by a single entity such as a large organization. Less than 100 km.

Explanation:

A **Metropolitan Area Network (MAN)** typically spans a city or a campus and covers a range of less than 100 kilometers. It is larger than a Local Area Network (LAN) but smaller than a Wide Area Network (WAN). MANs are commonly used by organizations or municipalities to connect various locations within a city or metropolitan area.

The subnet mask for **Gigabit0/0** on **Router0 (R0)** in **Packet Tracer (PT)** depends on the IP addressing configuration used in your specific lab setup. However, the most common subnet mask for a standard network configuration is:

255.255.255.0

Explanation:

- **255.255.255.0** is a common subnet mask for Class C networks, which allows for 256 IP addresses (254 usable for hosts).
- Unless specified otherwise, lab exercises often use this mask for simplicity in small networks.

If you're unsure, open **Router0 (R0)** in Packet Tracer, go to the **CLI** or **Config** tab, and check the configuration for **Gigabit0/0** to confirm.

The items considered **media** are:

- **Coaxial**
- **Wireless**

Explanation:

- **Coaxial**: A type of physical transmission medium used to carry signals (e.g., cable internet, TV signals).
- **Wireless**: A non-physical transmission medium that uses radio waves or other wireless signals for communication.

Not Media:

- **RJ45**: This is a connector used for Ethernet cables, not a medium itself.
- **Software games**: These are applications and not related to transmission media.

The correct answer is:

Straight-through cable

Explanation:

- In Packet Tracer **Lab 1**, when connecting a **PC (PC1)** to a **Switch (Switch1)**, a **straight-through cable** is typically used. This is because straight-through cables are used to connect different types of devices, such as a PC to a switch or a switch to a router.

Why not the other options?

- **RJ45 cable**: This is a connector type, not a cable type. Straight-through cables often have RJ45 connectors at both ends.
- **Crossover cable**: These are used to connect similar devices, such as PC-to-PC or switch-to-switch.
- **Coaxial cable**: This is not used in modern Ethernet networks; it's used for TV or broadband internet in older systems.

The correct answer is:

ipconfig /all

Explanation:

- The `**ipconfig /all**` command displays detailed network configuration information for a PC, including:
  - **Default Gateway**: The IP address of the router or gateway the PC uses to communicate outside its local network.
  - **MAC Address**: Also called the physical address, it uniquely identifies the network interface card (NIC) of the PC.

Why not the other options?

- `**ipconfig /DHCP**`: This is not a valid command.
- `**ipconfig / Ping**`: This is incorrect and doesn't exist as a valid command combination.
- `**ipconfig /flashdns**`: This clears the DNS resolver cache but does not display the gateway or MAC address.

The correct answer is:

Ping

Explanation:

- The `**ping**` command is used to test connectivity between two devices on a network. It sends ICMP (Internet Control Message Protocol) echo requests to the destination device and waits for a reply, confirming that the two devices can communicate.

Why not the other options?

- `**ipconfig**`: Displays IP configuration details but does not test connectivity.
- `**ipconfig /all**`: Provides detailed configuration information but does not test connectivity.
- `**Tracer**`: This is not a valid command; the correct tool for tracing the route is `**tracert**` or `**traceroute**`, but it is not typically used to test basic connectivity.

The correct answer is:

Mesh

Explanation:

- The **Internet** resembles a **mesh topology** because it involves multiple interconnected devices (routers, servers, and nodes) that communicate with one another directly or indirectly.
- In a mesh topology, devices are connected in such a way that data can travel through multiple paths, ensuring redundancy and fault tolerance, which are key features of the Internet's design.

Why not the other options?

- **Bus**: In a bus topology, all devices are connected to a single central cable, which is not scalable or fault-tolerant for the Internet.
- **Ring**: In a ring topology, each device connects to exactly two others to form a circular path, which is not representative of the Internet's structure.
- **Star**: In a star topology, all devices are connected to a central hub, which would create a single point of failure—not ideal for the Internet.

The correct answer is:

Crossover cable

Explanation:

- A **crossover cable** is used to connect two similar devices directly, such as **router-to-router**, **switch-to-switch**, or **PC-to-PC**. It swaps the transmit and receive pins to allow communication between the devices.

Why not the other options?

- **Straight-through cable**: Used to connect dissimilar devices, like a PC to a switch or a switch to a router.
- **Rollover cable**: Used to connect to a router or switch console port for configuration, not for communication between routers.
- **None of the above cable**: This is incorrect because the correct cable is a crossover cable.

The correct answer is:

Rollover cable

Explanation:

- A **rollover cable** (also called a console cable) is used to connect a PC to a router's **console port** for configuration purposes. This type of cable has one end with a reverse pinout compared to the other end.

Why not the other options?

- **Crossover cable**: Used to connect similar devices (e.g., router-to-router or PC-to-PC) but not for configuration purposes.
- **Straight-through cable**: Used to connect dissimilar devices (e.g., PC-to-switch or switch-to-router) but not for accessing the router console.
- **None of the above cable**: Incorrect because a rollover cable is the correct choice for this purpose.

The correct answers are:

Windows 2000
Linux Red Hat

Explanation:

An **NOS (Network Operating System)** is an operating system designed to manage network resources and provide services like file sharing, printing, and user management across a network.

- **Windows 2000**: Considered an NOS because it includes features like Active Directory, centralized user and resource management, and support for network services.
- **Linux Red Hat**: A popular NOS for servers, providing robust networking features, security, and support for enterprise-level network services.

Why not the others?

- **DOS**: Not a network operating system. It is a simple disk operating system with no native networking capabilities.
- **None of the above**: Incorrect because both Windows 2000 and Linux Red Hat are valid NOS examples.

True

Explanation:

- **UNIX (Red Hat)** is an example of a client/server network. In such networks, clients request services (e.g., file sharing, printing, web access), and servers provide those services. Red Hat Enterprise Linux is commonly used as a server operating system for hosting network services in client/server environments.

The correct answer is:

Star

Explanation:

- In a **Star topology**, all devices are connected to a central hub or switch. This central device acts as a point for all communication between devices, making it the hub of the network.

Why not the other options?

- **Bus**: In a bus topology, all devices are connected to a single central cable.
- **Ring**: In a ring topology, devices are connected in a circular path, not to a central hub.
- **Mesh**: In a mesh topology, devices are connected directly to each other, not through a central hub.

The correct answer is:

Star

Explanation:

- In a **Star topology**, all devices are connected to a central hub or switch. This hub or switch acts as the central point for communication between devices.

Why not the other options?

- **Bus**: In a bus topology, devices are connected to a single central cable, not through a hub or switch.
- **Ring**: In a ring topology, devices are connected in a circular path, not to a central hub or switch.
- **Mesh**: In a mesh topology, devices are connected directly to each other without relying on a hub or switch.

True

Explanation:

- **UNIX (Red Hat)** operating system is commonly used in client/server networks. It provides the necessary infrastructure for managing servers and handling client requests, making it a key component in client/server environments.

The correct answer is:

MAN

Explanation:

- **MAN (Metropolitan Area Network)** typically spans a city or a large campus and is typically operated by a single entity such as a large organization. It covers an area less than 100 kilometers.

Why not the other options?

- **WAN (Wide Area Network)**: Covers a larger geographical area, often spanning regions, countries, or even continents, and is typically not limited to less than 100 km.
- **LAN (Local Area Network)**: Operates within a small geographical area, such as an office or building, and typically has a range of less than 1 km.
- **SAN (Storage Area Network)**: Primarily deals with storage devices and does not directly cover geographic areas like MAN or LAN.

The correct answer is:

Somebody stole a Terminator from the network.

Explanation:

- In a **Bus (thin-net) topology**, devices are connected to a single central cable (coaxial cable), and each end of the cable must have a **Terminator** to prevent signal reflection. If a Terminator is removed or stolen, the network will fail to function properly because the signal cannot properly terminate, leading to communication issues.

The correct answer is:

It is a physical layout of your network.

Explanation:

- **Topology** refers to the physical or logical layout of a network. It defines how devices in a network are connected to each other, whether physically (e.g., through cables or wireless connections) or logically (e.g., through routing paths).

Why not the other options?

- **It is a physical MAC address of your network**: This refers to a specific device's MAC address, not the overall layout.
- **It is the logical IP address of your network**: While IP addresses are important for network communication, topology specifically refers to the layout or arrangement of the network, not just IP addresses.

The correct answer is:

You need a media to set up the network.

Explanation:

- To set up a network, at a minimum, you need a **media** (physical or wireless) to connect devices. This could be Ethernet cables, wireless signals, or other communication mediums.

Why not the other options?

- **Two people want to share**: While people are necessary for using the network, it's not a technical requirement.
- **The operating system should be NOS**: While a Network Operating System (NOS) is important for managing network resources, a media connection is the fundamental requirement.

The correct answer is:

All the above

Explanation:

- **Ethernet** can use various topologies such as bus, star, ring, and mesh.
- **Ethernet** sends broadcast signals to all devices on the network to communicate.
- **Ethernet** uses various hardware such as hubs, switches, RJ45 connectors, and in some cases, coaxial cables for communication.

Thus, all the listed characteristics apply to Ethernet.

In **Lab 1 (PT)**, if we consider all connections (cables) between devices such as PCs, switches, and routers, the total number is:

5

Explanation:

- Typically, in a basic network setup like Lab 1, you might have connections between:
    1. PC1 to Switch1
    2. Switch1 to Router0
    3. Router0 to another switch or hub (if present)
    4. Switch1 to PC2
    5. PC2 to another device or switch (depending on the setup).

Thus, **5** connections are commonly seen.

The correct option is:

Option, Tools, device name label

Explanation:

- In **Packet Tracer (PT)**, to add device name labels, you would typically use:
  - **Option**
  - **Tools**
  - Then select **Device Name Label** to add labels to devices.

Thus, the correct sequence is **Option, Tools, device name label**.

The correct answer is:

Because router 1941 by default does not have any interfaces

Explanation:

- By default, the Cisco **1941 Router** does not come with built-in interfaces (like Ethernet or Serial ports). Therefore, modules like **HWIC-2T** (which provides Fast Ethernet or Serial interfaces) need to be added to provide the necessary network interfaces for connectivity.

Thus, the HWIC-2T modules are added to provide additional interfaces for network connections.